SC4001 Neural Networks and Deep Learning

Group Project

Last update: 10 October 2026

Start date: 12 October 2026 Deadline: 16 November 2026 (11:59 PM)

Project aims and expectations

In groups of no more than three members, investigate an application or research question in neural networks and deep learning. Choose a suggested direction or another relevant topic that is feasible within the available time and resources.

Define a focused question, implement a suitable baseline, and investigate a well-motivated modification, comparison, or hypothesis. Evaluate appropriately, keep test data separate from model development, and analyse limitations and failure cases. Well-supported negative results are also valuable.

Acknowledge existing code and pretrained models, identify your contribution, and ensure every member can explain the work. Investigate model behaviour or methods through controlled experiments; assembling an application from existing tools alone is insufficient.

Submission requirements

Submit a PDF report and associated code in a ZIP file to NTULearn by the deadline. The report must use A4 pages, a single column, and Arial 10-point font, with a maximum of 10 pages excluding references, the contents page, the cover page, and appendix. Note that we expect all main content to be included within the ten pages and treat the appendix as optional material to examine.

Include team members' names and group number on the front page, an introduction, a review of existing techniques, methods, experiments and results, discussion, and an AI Use Statement. Comment your code and include instructions for reproducing the main results, covering dependencies, data preparation, and commands.

Assessment

Experiments and results: 40% | Report presentation: 30% | Novelty: 20% | Peer review: 10%

Novelty may include a justified methodological change, adaptation to a problem, or original investigation producing useful insights. A larger model or higher score alone does not establish a meaningful contribution.

The same late-submission penalty as in Assignment 1 applies: 5% for each day up to three days.

Using generative AI

You are encouraged, but not required, to use generative AI to develop ideas, explore methods, assist with coding and debugging, and improve writing. Verify claims and references against original sources, test

generated code, and report only experiments your team actually performed. You remain responsible for the submitted work. Hallucinated content will lead to mark deduction.

Include an AI Use Statement naming the tools and their purposes, how you checked or revised their outputs, and your team's decisions and contributions. If no generative AI was used, state this explicitly. Marks will be deducted if this declaration is omitted.

AI Use Statement example

Example, where accurate: "We used [tool and model] for brainstorming, debugging, and editing. We verified references and tested code. Our team chose the question, designed and ran experiments, and interpreted results."

Computational resources

Choose a manageable dataset and model size. Training a foundation model from scratch is not expected. You may use the computational resources assigned by the course or Google Colab. Plan for session interruptions and save checkpoints regularly.

Suggested project directions

These topics are starting points. Use generative AI to explore relevant literature, develop ideas, and refine a feasible research question, while verifying its suggestions. Build on existing models where appropriate and support your conclusions through controlled experiments. You are allowed to propose your own topic.

A. ECG heartbeat classification

Investigate deep learning for classifying heartbeats from ECG signals. Formulate a question about an aspect such as class imbalance, robustness, or interpretability, and evaluate it using an appropriate dataset.

B. Protein secondary structure prediction

Investigate the prediction of protein secondary structure from amino acid sequences. Choose a focused question concerning prediction quality, sequence representation, or generalisation.

C. Sentiment analysis

Investigate sentiment classification under a practical challenge, such as limited labelled data or a change of domain. Develop and evaluate an approach that addresses your chosen challenge.

D. Image captioning

Investigate a deep learning approach to generating descriptions of images. Define a focused question about caption quality, reliability, or computational efficiency, and evaluate the resulting captions critically.

E. Hallucination and Reliability in Vision Language Models

Investigate when vision language models describe objects, attributes, or relationships that are not supported by an image. Develop and evaluate an approach to improve visual grounding, recognise uncertainty, or reduce unsupported responses.

F. Reliable Retrieval Augmented Generation

Investigate how a language model answers questions using external documents when the retrieved evidence is incomplete, irrelevant, or contradictory. Develop and evaluate a method to improve evidence selection, answer faithfulness, or the ability to abstain when evidence is insufficient.

G. Efficient Reasoning with Small Language Models

Investigate how to improve the reasoning ability of a small, openly available language model under a fixed computational budget. Explore a focused approach involving adaptation, distillation, or inference strategies, and evaluate the trade-off between answer accuracy and computational cost.

H. Controllable and Personalised Image Generation

Investigate how to adapt or control a pretrained diffusion model to satisfy a particular visual requirement, such as preserving a subject's identity or following a spatial constraint. Evaluate both the desired control and its effects on image quality, diversity, and prompt adherence.

I. Adapting Foundation Models to Unseen Domains

Investigate how a pretrained vision or vision language model behaves when deployed on data that differ from its training distribution. Develop and evaluate an adaptation strategy using limited or unlabelled target data, examining both improvements and cases where adaptation harms performance.

J. Detecting AI Generated Images Across Generators

Investigate whether an image detector can distinguish real and generated images when the generator used at test time was not encountered during training. Develop and evaluate an approach that improves generalisation, including robustness to transformations such as compression or resizing.